

Chapter 9

Transformations

Unit 3 — Quadrilaterals and Circles

McGraw Hill Glencoe Geometry Texas, 2015

Key Vocabulary

Term	Definition
Transformation	Movement of a figure in a plane
Preimage	Original figure before transformation
Image	Resulting figure after transformation
Isometry	Transformation that preserves size and shape (congruence)
Rigid motion	Another name for isometry
Composition	Two or more transformations applied in sequence

Types of Transformations

Type	Isometry?	Preserves
Translation	✓	Size, shape, orientation
Reflection	✓	Size, shape; flips orientation
Rotation	✓	Size, shape, orientation
Dilation	✗	Shape (not size)

Translation

A **translation** moves every point the same distance and direction.

Coordinate notation: $(x, y) \rightarrow (x + a, y + b)$

Vector notation: $\langle a, b \rangle$ means move a units right and b units up.

Example: Translate $A(3, 2)$ by $\langle -1, 4 \rangle$:

$$A' = (3 + (-1), 2 + 4) = (2, 6)$$

Reflection

A **reflection** flips a figure over a line (the **line of reflection**).

Line of Reflection	Rule
x -axis	$(x, y) \rightarrow (x, -y)$
y -axis	$(x, y) \rightarrow (-x, y)$
$y = x$	$(x, y) \rightarrow (y, x)$
$y = -x$	$(x, y) \rightarrow (-y, -x)$

Rotation

A **rotation** turns a figure about a fixed **center point** by a given **angle**.

Common rotations about the origin:

Rotation	Rule
90° CCW	$(x, y) \rightarrow (-y, x)$
180°	$(x, y) \rightarrow (-x, -y)$
270° CCW (= 90° CW)	$(x, y) \rightarrow (y, -x)$
360°	$(x, y) \rightarrow (x, y)$

Example — Rotation 90° CCW

Rotate $P(4, -2)$ by 90° counterclockwise about the origin.

$$(x, y) \rightarrow (-y, x)$$

$$P(4, -2) \rightarrow P'(2, 4)$$

Theorem 9-1: Reflections Preserve Congruence

A reflection is an **isometry**: the image is congruent to the preimage.

$$\triangle ABC \cong \triangle A'B'C'$$

Theorem 9-2: Composition of Reflections — Parallel Lines

A composition of reflections over two **parallel** lines is equivalent to a **translation**.

The translation distance = **twice** the distance between the lines.

$$\text{distance of translation} = 2d(\ell_1, \ell_2)$$

Theorem 9-3: Composition of Reflections — Intersecting Lines

A composition of reflections over two **intersecting** lines is equivalent to a **rotation**.

The rotation angle = **twice** the angle between the lines.

$$\text{angle of rotation} = 2 \times \theta$$

Dilation

A **dilation** with center C and scale factor k maps each point P to P' such that:

$$CP' = k \cdot CP$$

Coordinate rule (center at origin):

$$(x, y) \rightarrow (kx, ky)$$

- $k > 1$: **enlargement**
- $0 < k < 1$: **reduction**
- $k = 1$: identity (no change)

Example — Dilation

Dilate $\triangle ABC$ with vertices $A(2, 4)$, $B(0, 2)$, $C(4, 0)$ by scale factor $k = 3$.

$$A' = (6, 12), \quad B' = (0, 6), \quad C' = (12, 0)$$

The image is **3 times larger** and **similar** to the original.

Symmetry

Type	Description
Line symmetry	Figure maps onto itself when reflected over a line
Rotational symmetry	Figure maps onto itself when rotated $< 360^\circ$
Point symmetry	Maps onto itself when rotated 180° about a center

Example: A regular hexagon has 6 lines of symmetry and rotational symmetry of order 6 (every 60°).

Chapter 9 — Transformation Summary

Transformation	Rule (origin)	Isometry
Translation $\langle a, b \rangle$	$(x + a, y + b)$	✓
Reflection over x -axis	$(x, -y)$	✓
Reflection over y -axis	$(-x, y)$	✓
Rotation 90° CCW	$(-y, x)$	✓
Rotation 180°	$(-x, -y)$	✓
Rotation 270° CCW	$(y, -x)$	✓
Dilation scale k	(kx, ky)	✗

Composition Theorems

Theorem	Condition	Equivalent To
Thm 9-2	Reflect over 2 parallel lines	Translation ($2\times$ distance)
Thm 9-3	Reflect over 2 intersecting lines	Rotation ($2\times$ angle)